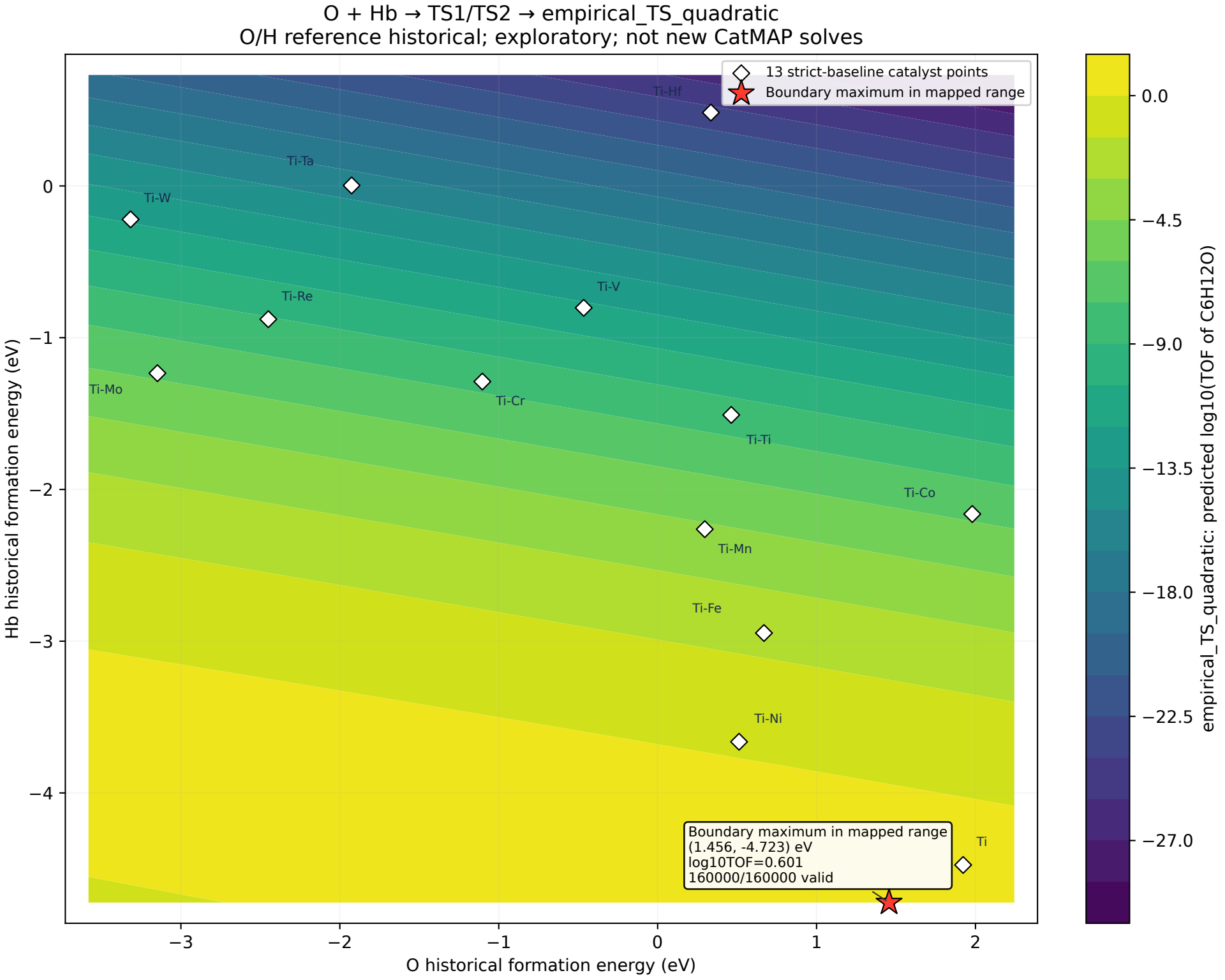

O + Hb → TS1/TS2 → empirical_TS_quadratic
O/H reference historical; exploratory; not new CatMAP solves
13 strict-baseline catalyst points
Boundary maximum in mapped range
Ti-Hf
Ti-Ta
Ti-W
Ti-Re
Ti-V
Ti-Mo
Ti-Cr
Ti-Ti
Ti-Co
Ti-Mn
Ti-Fe
Ti-Ni
Ti
Boundary maximum in mapped range
(1.456, -4.723) eV
log10TOF=0.601
160000/160000 valid
Hb historical formation energy (eV)
O historical formation energy (eV)
empirical_TS_quadratic: predicted log10(TOF of C6H12O)
0
−1
−2
−3
−4
1
2
0.0
−4.5
−9.0
−13.5
−18.0
−22.5
−27.0